

# Complete E-Learning Platform System Handover Documentation

**Project:** AI-Powered E-Learning Platform with Market Intelligence

**Date:** September 10, 2025

**Platform:** Raspberry Pi 4 (8GB RAM)

**Domain:** <https://elearning.wiabtech.in>

**Status:** Production Ready - Full Stack Operational

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## System Overview

### Vision & Purpose

A comprehensive AI-powered e-learning platform that provides:

- Data-driven course development based on real job market intelligence

- Interactive learning experiences with multimedia content
- Advanced admin dashboard with market insights
- External API integration for business intelligence
- Scalable architecture for educational content delivery

## Key Achievements

- **Production-ready e-learning platform** serving courses and managing users
  - **External market intelligence API** providing real-time job market data
  - **Cloudflare Tunnel integration** for secure external access
  - **Automated monitoring and health management**
  - **Mobile-responsive design** with PWA capabilities
  - **Real-time business intelligence** dashboard for course development strategy
- 

## Hardware Architecture

### Primary Server

#### Raspberry Pi 4 Model B

- **RAM:** 8GB LPDDR4-3200
- **Storage:** 64GB+ MicroSD (Class 10, A2)
- **Network:** Gigabit Ethernet + 802.11ac WiFi
- **IP Address:** 192.168.1.216 (Static)
- **Operating System:** Ubuntu Server 24.04 LTS

### Performance Specifications

- **CPU:** Broadcom BCM2711, Quad-core Cortex-A72 @ 1.5GHz
- **GPU:** VideoCore VI
- **USB:** 2x USB 3.0, 2x USB 2.0
- **Power:** 5V 3A USB-C (Official Raspberry Pi PSU)
- **Cooling:** Passive heatsink + optional fan

### Storage Architecture

```
/home/roland/e_learning_portal/ # Main application directory
```

|                            |                              |
|----------------------------|------------------------------|
| — instance/                | # Database files             |
| — static/                  | # Static assets              |
| — templates/               | # HTML templates             |
| — content/                 | # Course content             |
| — job_market_intelligence/ | # Market intelligence system |
| — venv/                    | # Python virtual environment |

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## Network Infrastructure

### Domain Configuration

- **Primary Domain:** elearning.wiabtech.in
- **API Subdomain:** api.wiabtech.in
- **DNS Provider:** Cloudflare
- **SSL:** Universal SSL Certificate (Wildcard: \*.wiabtech.in)

### Cloudflare Tunnel Configuration

```
yaml
# ~/.cloudflared/config.yml
tunnel: 4cca3de3-cc66-48cf-865d-fbc6c58d4b01
credentials-file: /home/roland/.cloudflared/4cca3de3-cc66-48cf-865d-fbc6c58d4b01.json
ingress:
  - hostname: api.wiabtech.in
    service: http://localhost:5001
  - hostname: elearning.wiabtech.in
    service: http://192.168.1.216:8000
  - service: http_status:404
```

### Network Services

- **Main Application:** Port 8000 (Gunicorn)
  - **Market Intelligence API:** Port 5001 (Flask)
  - **Cloudflare Tunnel:** Automatic port management
  - **SSH:** Port 22 (Local network only)
-

# Core Application Stack

## Technology Stack

yaml

### Backend:

- Python 3.12.3
- Flask 3.0.x (Web Framework)
- Gunicorn (WSGI Server)
- SQLAlchemy (ORM)
- SQLite3 (Database)

### Frontend:

- Bootstrap 5.3.3 (CSS Framework)
- Font Awesome 6.4.0 (Icons)
- Vanilla JavaScript (Interactivity)
- Chart.js (Data Visualization)

### Infrastructure:

- Ubuntu Server 24.04 LTS
- Systemd (Service Management)
- Cloudflare Tunnel (External Access)
- Virtual Environment (Python Isolation)

## Main Application (app.py)

python

*# Core Components:*

- Flask application factory
- User authentication & session management
- Admin dashboard **with** market intelligence
- Course management system
- Content delivery system
- File upload handling
- Database models & relationships
- API endpoints **for** market intelligence
- Health monitoring integration

## Key Features Implemented

1. **User Management:** Registration, login, session timeout

- 2. **Course System:** Creation, editing, publishing, enrollment
  - 3. **Content Management:** Sections, pages, file uploads
  - 4. **Admin Dashboard:** Real-time statistics, market intelligence
  - 5. **Market Intelligence:** Job data integration, recommendations
  - 6. **Health Monitoring:** System metrics, performance tracking
  - 7. **File Management:** Cover images, content uploads, hero images
- 

## Market Intelligence System

### Architecture Overview



### Data Sources

- **Primary API:** Jooble.org (500 requests/day)
- **Data Coverage:** 36 tech skills in Indian job market
- **Update Frequency:** Manual/scheduled refresh
- **Data Quality:** Real job postings from multiple job boards

### Market Intelligence Components

#### 1. MarketIntelligenceClient (app.py)

```
python
```

```

class MarketIntelligenceClient:
    def __init__(self, api_base_url="https://api.wiabtech.in"):
        self.api_base_url = api_base_url

    def get_skills_data(self):
        # Fetch skills data from external API

    def get_course_recommendations(self):
        # Fetch course recommendations

    def get_market_summary(self):
        # Generate market summary statistics

```

## 2. API Endpoints

- `/admin/market-intelligence` - Full dashboard
- `/api/admin/market-summary` - Real-time summary
- `/api/admin/skills-by-priority/<priority>` - Filtered data
- `/api/admin/all-skills-data` - Complete overview
- `/api/admin/refresh-market-data` - Force refresh

## 3. Database Integration

```

sql
-- External API data structure
{
  "skills": [
    {
      "skill": "python developer",
      "demand": 30,
      "source": "Jooble.org"
    }
  ]
}

```

## Business Intelligence Categories

- High Priority (26 skills):** 20+ jobs each - Immediate development
- Niche Opportunities (4 skills):** 8-16 jobs - Specialized markets
- Low Priority (6 skills):** 0-4 jobs - Avoid development

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# File Structure & Architecture

## Complete Directory Structure

```
/home/roland/e_learning_portal/
├── app.py                # Main Flask application
├── models.py             # Database models
├── requirements.txt       # Python dependencies
├── .gitignore            # Git ignore rules
├── instance/
│   ├── portal.db         # Main SQLite database
│   └── market_intelligence.db # Market data (if separate)
├── static/
│   ├── css/
│   ├── js/
│   ├── images/
│   ├── uploads/
│   │   ├── covers/      # Course cover images
│   │   └── hero/        # Hero images
│   ├── manifest.json     # PWA manifest
│   └── sw.js             # Service worker
├── templates/
│   ├── base.html         # Base template
│   ├── index.html        # Landing page
│   ├── login.html        # Authentication
│   ├── register.html     # User registration
│   ├── catalog.html      # Course catalog
│   ├── admin_dashboard.html # Enhanced admin dashboard
│   ├── admin_market_intelligence.html # Market intelligence page
│   ├── admin_courses.html # Course management
│   ├── admin_server_health.html # System monitoring
│   └── [other admin templates]
├── content/
│   └── pages/            # Course content files
├── job_market_intelligence/
│   ├── api_endpoints.py  # Market intelligence API server
│   ├── jooble_api_scraper.py # Data collection
│   ├── job_market_intelligence.db # Market data database
│   └── [supporting scripts]
├── venv/                 # Python virtual environment
└── [development & testing files]
```

# Critical Files

## 1. Main Application (app.py)

- **Size:** ~2000+ lines
- **Functions:** 50+ routes and functions
- **Key Sections:**
  - Database configuration
  - User authentication
  - Admin dashboard
  - Course management
  - Market intelligence integration
  - Health monitoring
  - File upload handling

## 2. Database Models (models.py)

python

```
class User(UserMixin, db.Model):  
    # User authentication and profile  
  
class Course(db.Model):  
    # Course information and metadata  
  
class Section(db.Model):  
    # Course sections organization  
  
class Page(db.Model):  
    # Individual lesson pages  
  
class Enrollment(db.Model):  
    # User-course relationships  
  
class CourseFeedback(db.Model):  
    # User feedback system
```

## 3. Templates Architecture

- **Base Template:** Responsive design with dark mode

- **Admin Templates:** Dashboard, course management, analytics
  - **Public Templates:** Landing, catalog, course viewing
  - **Authentication:** Login, registration, password reset
- 

## Services & Automation

### Systemd Services

#### 1. E-Learning Portal Service

```
ini
# /etc/systemd/system/elearning-portal.service
[Unit]
Description=E-Learning Portal (Gunicorn)
After=network.target

[Service]
Type=simple
User=roland
WorkingDirectory=/home/roland/e_learning_portal
Environment=PATH=/home/roland/e_learning_portal/venv/bin
ExecStart=/home/roland/e_learning_portal/venv/bin/gunicorn -w 4 -k gthread --threads 8 --bind 192.168.1.216:8000 a
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

#### 2. Market Intelligence API Service

```
ini
```

```
# /etc/systemd/system/job-market-api.service
[Unit]
Description=Job Market Intelligence API Server
After=network.target

[Service]
Type=simple
User=roland
WorkingDirectory=/home/roland/e_learning_portal/job_market_intelligence
Environment=PATH=/home/roland/e_learning_portal/venv/bin
ExecStart=/home/roland/e_learning_portal/venv/bin/python3 -c "from api_endpoints import create_job_market_app; a
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

### 3. Cloudflare Tunnel Service

```
ini

# /etc/systemd/system/cloudflared.service
[Unit]
Description=Cloudflare Tunnel (elearning-pi)
After=network.target

[Service]
Type=simple
User=roland
ExecStart=/usr/bin/cloudflared tunnel --config /home/roland/.cloudflared/config.yml run
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

### Service Management Commands

```
bash
```

*# Status checking*

`sudo systemctl status elearning-portal`

`sudo systemctl status job-market-api`

`sudo systemctl status cloudflared`

*# Service control*

`sudo systemctl start|stop|restart [service-name]`

`sudo systemctl enable|disable [service-name]`

*# Log monitoring*

`sudo journalctl -u elearning-portal -f`

`sudo journalctl -u job-market-api --lines=50`

## Automation Scripts

### 1. Health Monitoring (Integrated in app.py)

python

`def get_system_metrics():`

*# CPU, memory, disk usage monitoring*

`def get_database_health():`

*# Database connectivity and performance*

`def get_health_status():`

*# Overall system health assessment*

### 2. Market Data Collection

python

*# job\_market\_intelligence/jooble\_api\_scraper.py*

- Automated job data collection **from** Jooble API

- Rate limiting compliance (**500** requests/day)

- Data validation **and** storage

- Error handling **and** retry logic

# Database Architecture

## Primary Database (portal.db)

```
sql

-- Core Tables
Users          # User accounts and authentication
Courses        # Course metadata and settings
Sections       # Course organization structure
Pages          # Individual lesson content
Enrollments    # User-course relationships
CourseFeedback # User feedback and ratings
UserActivity   # Session and activity tracking

-- Relationships
Users 1:N Enrollments N:1 Courses
Courses 1:N Sections 1:N Pages
Users 1:N CourseFeedback N:1 Courses
```

## Market Intelligence Database

```
sql

-- External API Data Cache
india_job_data # Job market statistics
skill_categories # Skill classification
trend_analysis # Historical data tracking
```

## Database Configuration

```
python

# SQLAlchemy Configuration
SQLALCHEMY_DATABASE_URI = f'sqlite:///{{DB_PATH}}'
SQLALCHEMY_TRACK_MODIFICATIONS = False
SQLALCHEMY_BINDS = {
    'trends': f'sqlite:///{{MARKET_INTELLIGENCE_DB_PATH}}'
}
```

---

# Security & Authentication

## User Authentication

- **Library:** Flask-Login
- **Session Management:** 10-minute timeout
- **Password Hashing:** Werkzeug security
- **Admin Protection:** Role-based access control

## Security Features

```
python

@login_required
@admin_required
def admin_routes():
    # Protected admin functionality

# Session timeout management
@app.before_request
def check_session_timeout():
    # Automatic logout on inactivity
```

## Data Protection

- **Database:** Local SQLite (no external exposure)
- **File Uploads:** Secure filename handling
- **API Keys:** Environment-based configuration
- **SSL/TLS:** Cloudflare Universal SSL

## Access Control

- **Local Network:** Direct IP access (192.168.1.216)
- **External Access:** Cloudflare Tunnel with SSL
- **Admin Interface:** Authentication required
- **API Endpoints:** Admin role verification

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## API Architecture

### Internal APIs

## 1. Admin Dashboard APIs

```
python
```

```
# Real-time data endpoints
```

```
/api/admin/market-summary           # Market intelligence summary
```

```
/api/admin/skills-by-priority/<priority>  # Filtered skills data
```

```
/api/admin/all-skills-data           # Complete skills overview
```

```
/api/admin/refresh-market-data       # Force data refresh
```

## 2. System Health APIs

```
python
```

```
# Monitoring endpoints
```

```
/admin/server-health                # Health dashboard
```

```
/api/system/metrics                # System metrics (if implemented)
```

## External API Integration

### 1. Jooble.org API

```
python
```

```
# Configuration
```

```
API_KEY = "c27d782e-8593-4bcc-b7b4-42b7d36e04db"
```

```
BASE_URL = "https://jooble.org/api/"
```

```
RATE_LIMIT = 500 # requests per day
```

```
# Request format
```

```
{  
    "keywords": "python developer",  
    "location": "India"  
}
```

### 2. Market Intelligence API (api.wiabtech.in)

```
python
```

```
# External endpoints
```

```
https://api.wiabtech.in/           # API status
```

```
https://api.wiabtech.in/api/job-market/skills  # Skills data
```

```
https://api.wiabtech.in/api/job-market/course-recommendations # Recommendations
```

---

# Monitoring & Health

## System Monitoring Features

- 1. **Real-time Metrics:** CPU, memory, disk usage
- 2. **Service Health:** Application status monitoring
- 3. **Database Health:** Connection and performance metrics
- 4. **API Monitoring:** External service availability
- 5. **Error Tracking:** Application error logging

## Health Dashboard Components

```
python

# System metrics collection
CPU Usage, Memory Usage, Disk Space
Network Status, Service Status
Database Connectivity, API Response Times
```

## Alerting (Future Implementation)

- Email notifications for critical issues
- Automated service restart on failure
- Performance threshold monitoring
- External API outage detection

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# Deployment & Operations

## Production Deployment Process

### 1. Environment Setup

```
bash
```

```
# System preparation
sudo apt update && sudo apt upgrade
sudo apt install python3-pip python3-venv nginx

# Application setup
cd /home/roland
git clone [repository-url] e_learning_portal
cd e_learning_portal
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

## 2. Database Initialization

```
bash

# Create databases
python3 -c "from app import db; db.create_all()"

# Verify setup
sqlite3 instance/portal.db ".tables"
```

## 3. Service Configuration

```
bash

# Copy service files
sudo cp [service-files] /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable elearning-portal job-market-api cloudflared
sudo systemctl start elearning-portal job-market-api cloudflared
```

## 4. Cloudflare Tunnel Setup

```
bash

# Install cloudflared
wget [cloudflared-url]
sudo mv cloudflared /usr/local/bin/
cloudflared tunnel login
cloudflared tunnel create elearning-pi
# Configure and start tunnel
```

## Backup Procedures

```
bash

# Database backup
cp instance/portal.db backups/portal_$(date +%Y%m%d_%H%M%S).db

# Configuration backup
tar -czf config_backup.tar.gz ~/.cloudflared/

# Full application backup
rsync -av /home/roland/e_learning_portal/ /backup/location/
```

## Update Procedures

```
bash

# Application updates
cd /home/roland/e_learning_portal
git pull origin main
source venv/bin/activate
pip install -r requirements.txt
sudo systemctl restart elearning-portal job-market-api
```

---

## Development Workflow

### Git Workflow

```
bash

# Current branch structure
main                # Production ready code
feature/pwa-implementation  # Current development branch

# Development process
git checkout -b feature/new-feature
# Make changes
git add .
git commit -m "feat: description"
git push origin feature/new-feature
# Create pull request
```

## Development Environment

```
bash

# Local development
source venv/bin/activate
export FLASK_ENV=development
python app.py

# Testing
curl http://localhost:8000
curl http://localhost:5001/api/job-market/skills
```

## Code Quality Standards

- **Documentation:** Comprehensive docstrings
  - **Error Handling:** Try-catch blocks for external APIs
  - **Logging:** Structured logging for debugging
  - **Security:** Input validation and sanitization
- 

## Troubleshooting Guide

### Common Issues

#### 1. Service Not Starting

```
bash

# Check service status
sudo systemctl status [service-name]

# View logs
sudo journalctl -u [service-name] --lines=50

# Common fixes
sudo systemctl daemon-reload
sudo systemctl restart [service-name]
```

#### 2. Database Issues

```
bash
```

```
# Check database file permissions
```

```
ls -la instance/portal.db
```

```
# Test database connectivity
```

```
python3 -c "from app import db; print(db.engine.execute('SELECT 1').scalar())"
```

```
# Database repair (if needed)
```

```
sqlite3 instance/portal.db ".backup main backup.db"
```

### 3. API Connectivity Issues

```
bash
```

```
# Test internal APIs
```

```
curl http://localhost:8000/admin
```

```
curl http://localhost:5001/api/job-market/skills
```

```
# Test external access
```

```
curl https://elearning.wiabtech.in/
```

```
curl https://api.wiabtech.in/api/job-market/skills
```

```
# Check Cloudflare tunnel
```

```
sudo systemctl status cloudflared
```

```
cloudflared tunnel info elearning-pi
```

### 4. Market Intelligence Issues

```
bash
```

```
# Test Jooble API directly
```

```
curl -X POST "https://jooble.org/api/c27d782e-8593-4bcc-b7b4-42b7d36e04db" \  
-H "Content-Type: application/json" \  
-d '{"keywords": "python developer", "location": "India"}'
```

```
# Check API rate limits
```

```
# Monitor usage in application logs
```

```
# Refresh market data
```

```
curl http://localhost:8000/api/admin/refresh-market-data
```

## Recovery Procedures

### Emergency Recovery

```
bash
```

```
# Stop all services
```

```
sudo systemctl stop elearning-portal job-market-api
```

```
# Restore from backup
```

```
cp backups/portal_latest.db instance/portal.db
```

```
# Restart services
```

```
sudo systemctl start elearning-portal job-market-api
```

```
# Verify functionality
```

```
curl http://localhost:8000/admin
```

## Complete System Rebuild

```
bash
```

```
# Backup critical data
```

```
cp -r instance/ ~/backup_instance/
```

```
cp -r content/ ~/backup_content/
```

```
# Fresh installation
```

```
rm -rf e_learning_portal/
```

```
git clone [repository] e_learning_portal
```

```
# Follow deployment process
```

```
# Restore data from backups
```

---

## Business Intelligence

### Market Intelligence Insights

#### Current Market Analysis (September 2025)

- **Total Skills Tracked:** 36
- **High Priority Skills:** 26 (immediate development recommended)
- **Niche Opportunities:** 4 (specialized market potential)
- **Low Priority Skills:** 6 (oversaturated markets)

#### Top Demand Skills (30+ jobs each)

1. Python Developer
2. Java Developer
3. JavaScript Developer
4. React Developer
5. Data Scientist
6. Machine Learning Engineer
7. UI/UX Designer
8. Business Analyst

### **Skills to Avoid (0-4 jobs)**

1. Blockchain Developer (0 jobs)
2. Jenkins (4 jobs)
3. Azure Developer (2 jobs)
4. Docker (2 jobs)
5. Terraform (2 jobs)
6. Kotlin Developer (2 jobs)

### **Course Development Strategy**

- **Immediate Priority:** Develop courses for 30+ demand skills
- **Strategic Planning:** Consider niche opportunities for specialized audiences
- **Resource Allocation:** Avoid course development in oversaturated areas
- **Market Monitoring:** Regular updates to track demand changes

### **Revenue Impact Analysis**

- **High ROI Potential:** Python, Java, JavaScript courses
- **Specialized Markets:** AI/ML, Cybersecurity training
- **Cost Avoidance:** Not developing blockchain courses saves resources

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## **Future Roadmap**

### **Immediate Enhancements (1-2 weeks)**

1. **Automated Data Collection:** Schedule daily Jooble API updates

2. **Enhanced Analytics:** Historical trend tracking
3. **Mobile Optimization:** PWA feature completion
4. **Performance Optimization:** Caching mechanisms

### Short-term Development (1-3 months)

1. **Advanced Dashboard:** Predictive analytics
2. **API Expansion:** Additional data sources integration
3. **Automation:** Smart course recommendations
4. **User Experience:** Enhanced admin interface

### Long-term Vision (3-6 months)

1. **Machine Learning:** Demand forecasting models
2. **Multi-tenant Architecture:** Support multiple institutions
3. **Advanced Analytics:** Student outcome correlation
4. **Mobile Application:** Native mobile app development

### Scalability Considerations

1. **Database Migration:** PostgreSQL for production scale
  2. **Containerization:** Docker deployment
  3. **Load Balancing:** Multi-instance deployment
  4. **CDN Integration:** Global content delivery
- 

## Technical Specifications

### System Requirements

- **Minimum RAM:** 4GB (8GB recommended)
- **Storage:** 32GB minimum (64GB+ recommended)
- **Network:** Stable internet connection
- **Power:** Reliable power supply with UPS backup

### Performance Benchmarks

- **Response Time:** <500ms for dashboard pages
- **API Performance:** <200ms for market data endpoints

- **Concurrent Users:** 50+ supported on current hardware
- **Uptime Target:** 99.5% availability

## Monitoring Metrics

- **System Load:** CPU < 70%, Memory < 80%
  - **Database Performance:** Query time < 100ms
  - **External API:** Response time < 2s
  - **Storage:** Disk usage < 85%
- 

## Contact & Support

### System Administration

- **Primary Admin:** [roland@wiabtech.in](mailto:roland@wiabtech.in)
- **Server Location:** Local network (192.168.1.216)
- **External Access:** <https://elearning.wiabtech.in>
- **API Access:** <https://api.wiabtech.in>

### Key Resources

- **GitHub Repository:** [Your repository URL]
- **Documentation:** This handover document
- **API Documentation:** Integrated in dashboard
- **Support:** Admin dashboard health monitoring

### Emergency Contacts

- **Technical Issues:** Check systemd logs
  - **API Issues:** Monitor Cloudflare tunnel status
  - **Database Issues:** Backup restoration procedures
  - **External Dependencies:** Jooble.org API status
- 

## Conclusion

This e-learning platform represents a complete educational technology solution with advanced market intelligence capabilities. The system provides:

- **Production-ready infrastructure** on cost-effective hardware
- **Real-time business intelligence** for strategic course development
- **Scalable architecture** ready for expansion
- **Comprehensive monitoring** and health management
- **External API access** for future integrations

The platform successfully combines educational content delivery with data-driven business intelligence, providing a competitive advantage through market-informed course development strategies.

**System Status:** Production Ready

**Last Updated:** September 10, 2025

**Version:** 2.0.0 (Market Intelligence Integration Complete)

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*This document serves as the complete technical handover for the e-learning platform system. All components, services, and procedures are documented for seamless operation and future development.*